



The Least Squares Method in Multiple Regression Transcript

Hello everyone, welcome to lesson 1 video 2. In this video, we will learn how to estimate coefficients of multiple linear regression model using the least square method. We have learned that multiple regressions let us build powerful models with several predictors or explanatory variables, right? We know our model is no longer a simple line, but in 3D, it's a high dimensional or higher dimensional hyperplane, right? So that leads us to the next big question. How do we find that hyperplane? And the good news is, this is residing in something that we have learned much before in our school days or maybe some of you in your graduations.

But the principle that makes up all of these things is already known to you. And we are going to uncover all of those things in this session, okay? So the goal is to minimise sum of squared of errors to find out the best line or best hyperplane. This is also what you've learned in simple linear regression.

We are just extending that to multiple. So, whatever value that you predict, we try to find out how far it is from the actual value. And we try to find out or we try to find out coefficients or the combination of coefficients when with minimises the sum of squares.

So this is the equation that we have. This is a simple equation $\hat{y}$ is equals to β_0 plus $\beta_1 x_1$ plus $\beta_2 x_2$ plus and so on. And because you have all of that data, that's why if you see x_i , x_i , x_i , because you have n number of rows as well, right? You do not have only columns, which is x_1 , x_2 and so on.

Just like here, you have, let's say, price of the rooms, you have one column, you have a square foot, which is the input variable x_1 . This is your input variable x_2 . And you are trying to find out how does the price of houses changes because of the square fits or the dimension of the house and the bedrooms that it has, right? So let's get into the details of how do we get to sum of square of errors. So what do you have here? If you add the error term, because y does not only gets affected because of what you have included in the model, but there are random noise in the data as well, right? So you can add that term, you can take all of these betas or the coefficients along with the variables towards the left hand side, square it up and add it for all the variables or all the data points, then you will be getting this equation, which is sum of square of errors. So our goal is to find the coefficients that minimises this sum of square of errors, right? This becomes very tricky.

If you think of you have too many variables, you will have to use calculus, you will have to take the derivative of sum of square of errors with respect to each of these coefficients. And then you will get to equate them to zero, find that value where they are zero. And then you will substitute back solve multiple equations and get to the values of these coefficients, which is a tedious way of doing it.

So what we have, we have a simplified way of doing it using linear algebra or matrix algebra. So we use linear algebra and that's what I was telling you, majority of you have already gone through the topic which helps us get to coefficients, which is linear algebra. Using the matrices, we will be able to get to the equations or get to the final value using simple equation here, what we define it as here, right? You have x . Now what exactly is this x ? Think of this data, these two highlighted ones are your x variables.

Now because it is arranged in rows and columns, now this highlighted matrix is your x matrix. If you change square fit, so let's say for example, rows to columns and columns to rows, that becomes your x transpose. And that is what has been denoted there.

This is x transpose. And you multiply this x transpose, when you multiply this x transpose with x , takes inverse of that, okay? And then you multiply it again with x transpose and finally with y , you will get a vector, which is the coefficient vector. And that vector, in that vector, you would have all the coefficient values.

So in this particular case, let's say for example here, you might get a vector like price is equals to 100 times of bedrooms, okay? Or let's say for example, this is the coefficient of bedrooms. And you might get a square fit coefficient, which is 10 times of square fit, hence the coefficient of SQFT is this, right? So you will get a vector, which would look exactly like this. So you will have a vector, which is right now, because you have two dimensional data, it would be two cross one row vector.

You would have two rows, one column. In each row, each of those coefficients would be. So this is how it would look like.

This is how the entire B vector would look like. What do you have to do? This entire equation is derived using linear algebra or using calculus with vector, vector calculus basically. But you can easily derive that there are a lot of ways and if time permits, we will be going through it in the live class.

But if you think of, you have the X matrix, which is your input data I was showing you, right? This is your X matrix. What you do is you just take a transpose of it, multiply it with the same thing without transpose. And once you multiply it, take the inverse, you all, I think most of you would be knowing by this time or let's say most of you would have calculated at least inverses once you take the inverse of that matrix, the output matrix that you have got after multiplying X transpose with X and then whatever comes out of that, you multiply it again with X transpose.

Whatever comes out of that, you multiply it again with Y , you'll be getting a B vector. B vector or the elements of B vector would be those many number of rows as the number of dimensions that you have, or let's say if you X are X_1 and X_2 , then you would have two dimensional or let's say two rows in that B vector. Just like how I was showing you here, right? Because I had two dimensions coefficient, let's say bedrooms and square feet.



I have a two dimensional row vector, which is nothing but two cross one row vector. OK, so I hope this is all clear with everyone and I hope you are able to relate to this as well. In summary, the least square method finds the set of regression coefficients that best fit the observed data.

Although the calculations are more complex than in simple linear regression, the logic remains exactly same to minimise the square differences between predictor variables and the observed variable.

-----End-----